

LARRY HUI

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[LinkedIn](#) ◊ [GitHub](#) ◊ [Portfolio](#) ◊ [Google Scholar](#)

EDUCATION

University of Pennsylvania, GRASP Lab

May 2028 (*expected*)

M.S.E in Robotics, focus in AI, Control, and Perception
Cumulative GPA: XX/4.0

Relevant Coursework: World Models, Control and Optimization with Applications in Robotics, Machine Perception, Distributed Robotics

University of California, Berkeley

May 2026

B.S in Mechanical Engineering, focus on Electrical Engineering and Computer Science (EECS)
Cumulative GPA: 3.589/4.0; Major GPA: 3.659/4.0; Last Two Year GPA: 3.906/4.0

Relevant Coursework: Control for Bipedal Robotics; Convex Optimization; Machine Learning; Dynamics; Dynamical Systems & Feedback; Robotics; Electronics; Signals and Systems; Mechatronics Design; FEA

Graduate Coursework: Continuum Mechanics; Deep RL for Controls; Nonlinear Systems; Embedded Systems

Imperial College London

August 2023 — June 2024

B.S in Mechanical Engineering (UCEAP Study Abroad)

RESEARCH INTERESTS

Learning-based methods, control, and optimization for intelligent robotic systems; developing predictive and adaptive models that enable robots to perceive, plan, and act effectively in complex physical environments.

RESEARCH EXPERIENCE

Undergraduate Researcher, University of California Berkeley

December 2024 – May 2026

[Gu Research Group](#)

Berkeley, CA

- Formulated and solved PDEs in linear elasticity using a custom MATLAB/PyTorch PINN and MGN solver for MathWorks; reduced computational time by $\sim 30\%$ compared to baseline FEA for Timoshenko beam and a 2D central hole in plate using ensemble methods, a training pool with mesh metadata, and cosine annealing.
- Benchmarked PINN predictions against FEA data for several geometries and different BCs and tractions, achieving $< 5\%$ error in displacement fields preserving stability across 10k+ collocation points.
- Explored differentiable GNN-based solvers for fluid-solid interaction PDEs; implemented and normalized input and output features. Added a warm start for gradual training as well as batchnorm and layernorm.
- **Point-of-Contact:** Grace X. Gu (ggu@berkeley.edu)

Undergraduate Researcher, University of California Berkeley

January 2023 – May 2023

[FLOW Lab](#)

Berkeley, CA

- Incorporated principles of fluid-structure interaction and continuum mechanics into CAD modeling and experimental verification and testing of a Flow Loop Exhibit for the SF Sailing Science Centre.
- Served as a primary machinist, producing aluminum adapter plates to securely integrate 8 casters with 8020 framing using CNC mills; Engineered custom pump-support to enhance stiffness and dynamic stability.
- Optimized fixture assembly and applied DFMA principles, increasing fabrication efficiency by 30% while maintaining part clearances within 0.02" in tight assembly spaces for a splash proof & chilled X-ray detector.
- **Point-of-Contact:** Simo A. Mäkiharju (makiharju@berkeley.edu)

TECHNICAL EXPERIENCE AND PROJECTS

Pixel Mirror: Real-Time Multitasking and Virtual Gloving

January 2026 – May 2026

[Department of Mechanical Engineering, UC Berkeley](#)

Berkeley, CA

- Designed and assembled the hardware stack for an interactive digital art display, interfacing an ESP32 microcontroller with a 64×64 HUB75E LED matrix

- Developed the foundational CV silhouette tracking workflow using a YOLOv8-seg instance segmentation CNN, and engineered a high-performance native C++ port (`vision_fast.cpp`) to maximize frame processing efficiency.
- Programmed embedded C++ firmware utilizing I2S hardware peripherals in DMA mode to drive 4,096 RGB LEDs with minimal CPU overhead, implementing real-time button debouncing and EWMA signal smoothing for analog potentiometer inputs.
- Integrated the vision stack, MediaPipe hand tracking, and a multi-threaded PyQt6 GUI over a framed 1 Mbaud USB-CDC serial connection validation to guarantee an end-to-end latency under 70 ms at 30 fps.

Cable-Actuated Air Hockey Robot

January 2026 – May 2026

Department of Mechanical Engineering, UC Berkeley

Berkeley, CA

- Co-designed and manufactured a high-speed cable-driven robot utilizing 4 BLDC motors and an 80/20 aluminum frame, optimizing for low workspace inertia to achieve high-acceleration mallet positioning.
- Developed a real-time motion planning stack on a Jetson Nano executing an MPC strategy for defensive blocking and offensive strikes via continuous quintic-spline trajectory generation.
- Implemented an OpenCV computer vision pipeline with homography and parallax correction, fusing observations via an Extended Kalman Filter to track and predict puck trajectories with elastic wall-bounce reflections.
- Formulated closed-form inverse kinematics and an over-determined least-squares forward kinematics trilateration algorithm, streaming asynchronous, closed-loop motor commands to moteus controllers over a CAN bus.
- Engineered an auxiliary microcontroller firmware using a non-blocking UART protocol and memory-efficient parsers to handle flicker-free HMI rendering of live game trajectories on a TFT display.

Constrained RL for Safe Legged Locomotion over Transient Terrain

August 2025 – December 2025

Department of Mechanical Engineering, UC Berkeley

Berkeley, CA

- Developed a reinforcement learning framework for planar bipedal locomotion in OpenAI Gym, integrating Control Barrier/Lyapunov Functions (CBFs/CLFs) to ensure safety and prevent the agent from entering unstable configurations.
- Implemented multi-step look-ahead control to improve long-horizon decision-making on time-varying, stochastic discrete terrains, avoiding greedy one-step actions that compromise gait stability.
- Modeled hybrid locomotion dynamics with impact events and trained policies to generalize across terrains.

Autonomous Charger Insertion using the UR7e Robotic Arm

August 2025 – December 2025

Department of Electrical Engineering and Computer Science, UC Berkeley

Berkeley, CA

- Designed and implemented an autonomous cable-insertion system using the UR7e robotic arm with a custom 3D-printed end-effector and Intel RealSense depth sensing for precise cable and port localization
- Developed computer vision pipelines using OpenCV for multi-variant connector detection (USB-C, USB-A, Magsafe, Lightning) and inverse-kinematics motion planning for alignment and insertion in ROS2.
- Integrated sensing, planning, and actuation within a full perception-control stack to achieve centimeter accuracy and reliable insertion across randomized trials.
- Tuned force limits and implemented retry/re-grasp logic for compliance-based insertion and recovery from misalignment.

Finite Element Modeling and Simulation of Laser Processing

April 2025 – May 2025

Department of Mechanical Engineering, UC Berkeley

Berkeley, CA

- Implemented a 3D transient heat conduction solver in MATLAB to simulate laser heating on a cubic metallic block, modeling energy absorption via the Beer-Lambert law.
- Built a full FEM pipeline: mesh generation, numerical integration via Gaussian quadrature, shape function evaluation, and assembly of global stiffness and mass matrices with boundary-condition enforcement using the penalty method.
- Implemented Forward and Backward Euler schemes with both lumped and consistent mass matrices to advance temperature fields over time.
- Visualized transient temperature profiles at the top, mid, and bottom planes of the domain and analyzed timestep-dependent stability and accuracy effects.

PINNs for Fully Developed Heat Transfer in Non-Circular Cross-Sections

April 2025 – May 2025

Department of Mechanical Engineering, UC Berkeley

Berkeley, CA

- Developed and trained a PINN for fully developed laminar flow and convective heat transfer in rectangular channels.
- Designed a custom loss function combining physical residuals (Navier-Stokes and energy equations), boundary conditions, and target solutions to enforce physical consistency during training.

- Implemented network architecture and training in TensorFlow/Keras, tuning optimizers (Adam, SGD, AdamW) for convergence to loss < 0.3 over 10000 epochs.
- Quantified and visualized velocity and temperature fields; evaluated wall heat flux and compared to finite-difference benchmarks, revealing PINN overestimation due to coarse collocation sampling near boundary layers.
- Demonstrated that corner rounding in channels redistributes local heat flux, enhancing local cooling but reducing mean flux by $\sim 2\%$ and identified adaptive wall sampling as key to quantitative PINN accuracy.

Characterizing Thin Carbon Fibre Laminates under Quasi-Static Tension August 2024 – December 2024
Department of Mechanical Engineering, UC Berkeley *Berkeley, CA*

- Conducted quasi-static tensile testing on K13C2U carbon-fibre laminates for the Electron–Ion Collider detector project.
- Designed and fabricated stainless-steel and plywood wedge fixtures to enable ASTM D3039-compliant testing of sub-0.1 mm laminates using an Instron 6800 load frame.
- Applied Weibull and Kruskal–Wallis statistical analyses to quantify orientation-dependent tensile strength variability. Established Weibull moduli of 5.89 (transverse) and 4.39 (longitudinal), yielding composite moduli of 2.26 GPa and 3.33 GPa respectively.
- Demonstrated significant orientation effects on strength; validated through bootstrapped Monte Carlo sampling.

Hardware/Mechanical Engineering Intern July 2024 – December 2024
Outer Rim Exploration Inc. *Tokyo, Japan*

- Designed plastic CAD assembly for a drone-mounted muon hodoscope, integrating PCB layouts and validation systems for muon flux detection.
- Performed topology review, achieving a 10% mass reduction while increasing stiffness through geometric optimization and FEA verification via a custom MATLAB solver to analyze buckling and deflection under arbitrary conditions.
- Researched and compared deployable mechanism alternatives to linear slides—including pantographic, elastically deformable, and inflatable structures—to improve compactness and alignment stability.
- Designed a precision rotation stage using solenoid locking pins to control the hodoscope’s zenith angle with degree-level accuracy and geared bidirectional actuation.
- **Point-of-Contact:** Jeremiah Anderson (jeremiah.anderson@ore.green).

Compressed-Gas Car Prototype October 2023 – December 2023
Department of Mechanical Engineering, Imperial College London *London, UK*

- Co-designed and manufactured a compressed-gas-powered vehicle prototype using mills, lathes, 3D printers, CNCs, rollers, and laser cutters, applying Pugh’s Total Design methodology for concept evaluation.
- Engineered an encased Pelton turbine and shaft–pulley drivetrain optimized for high-torque energy transfer and minimal leakage losses.
- Produced toleranced engineering drawings and measurement reports; performed FEA to evaluate chassis stress, leading to the selection of Tensol-12 DCM and improving the factor of safety from 1.1 to 3.4.
- Managed the Gantt chart and design workflow, coordinating iterative testing and material selection across the team.

Financial Analyst Research Intern June 2022 – September 2022
RBC Dominion Securities Wealth Management *Toronto, ON*

- Assigned by the portfolio manager to research and evaluate the AI and technology sector for long-term investment potential, conducting industry and company-level financial analyses to forecast 5–10 year growth trends.
- Developed predictive analytics tools using LSTM neural networks and Plotly Dash dashboards to visualize performance trends and automate insight generation.
- Applied Modern Portfolio Theory (MPT) and quantitative risk models to evaluate asset performance, achieving a Sharpe Ratio of 3.1 for the optimized portfolio.

TEACHING

Lead Teaching Assistant, ME103: Experimentation and Measurements August 2025 – May 2026
Department of Mechanical Engineering, UC Berkeley *Berkeley, CA*

- Led teaching operations for 180+ for Junior and Senior students, coordinating 3 TAs and 3 Tutors/Readers and managing the Ed Forum with 30k+ interactions.
- Designed, graded, and implemented laboratory experiments and exams on signal processing, statistics, measurements, and experimental design using LabVIEW, MATLAB.

- Created/conducted labs, discussion sections, and office hours, supporting students in practical measurement and data analysis skills.

ME 375: Teaching of Mechanical Engineering at the University Level

August 2025 – Current

Department of Mechanical Engineering, UC Berkeley

Berkeley, CA

- Pedagogical seminar on educational theory, course design, inclusive learning, and lab instruction.
- Completed curriculum-development workshop reviewing the Mechanical Engineering undergraduate curriculum.
- Designed and presented syllabus improvements for ME 103 (Measurements & Experimentation), ME 132 (Dynamic Systems & Feedback), and ME 100 (Electronics for IoT) focusing on clear learning objectives, scaffolded lab progression, and integration of Gen AI and Guided-Discovery methods.
- Developed and led a Guided Discovery/Flipped Classroom activity for engineering instruction.

Lead Teaching Assistant, ME132: Dynamical Systems and Feedback Control

May 2025 – August 2025

Department of Mechanical Engineering, UC Berkeley

Berkeley, CA

- Led teaching operations for 80+ Dynamic Systems and Feedback Control for Juniors and Seniors, coordinating 1 TA and 2 Readers and managing the Ed Forum with 23.8k+ interactions.
- Developed and graded labs and exams, including Root Locus, Bode, Nyquist, controller design, and state-space problems emphasizing real-world applications.
- Led discussion sessions and office hours to clarify theoretical and computational concepts in MATLAB and Simulink.

Reader, MATH 52/1B: Calculus II

January 2025 – May 2025

Department of Mathematics, UC Berkeley

Berkeley, CA

- Managed Gradescope for 1000+ students and 2 discussion sections of 150 students. Graded weekly problem sets, quizzes, and exams with consistency and technical accuracy.

JOURNAL PUBLICATIONS (IN PREPARATION)¹

1. **L. Hui**, M. Tan, S Eleanor-Heiles, A. Kakde, A. Panggabean, K. Eyvazi, H. K. Taylor “*Statistical Reliability and Measurement Characterization of Ultra-Thin Carbon Fibre Laminates under Quasi-Static Tension*” Measurement Science and Technology (IOP Publishing), in preparation, 2025

CONFERENCE PUBLICATIONS

1. **L. Hui**, J. Kam, W. Su, and J. Zhou, “Hamilton–Jacobi Reachability for Spacecraft Collision Avoidance,” *2026 IEEE International Conference on Control & Automation (ICCA)*, 2026.

HONORS AND AWARDS

1. 2x Dean’s Honors List, College of Engineering (Top 10% GPA) — Fall 2025, Spring 2026

TECHNICAL SKILLS

Applications	SOLIDWORKS (CAD/FEA/CAM), Fusion360, OnShape, ANSYS, Simulink, GitHub
Design	AutoCAD, LabVIEW, ABAQUS, MS Office, Gmsh, Adobe Creative Cloud
Programming Languages	GD&T, Design for Manufacturing, Product Development, Design of Experiments
Frameworks/Libraries	Controller Design, Rapid Prototyping
Certifications	MATLAB, Python, Java, C++, Git, Linux, LabVIEW, L ^A T _E X
	ROS2, NumPy, SciPy, Pandas, TF, Matplotlib, Seaborn, PyTorch, OpenAI Gym
	NVIDIA Isaac Sim, CVX Matlab
	FAA/TC Private Pilot’s Licence (High-performance and Complex endorsements)
	ARCT Piano RCM, DELF B2

LANGUAGE SKILLS

Native Proficiency: English, Mandarin

Full Professional Proficiency: French (B2)

Elementary Proficiency: German (A1)

¹I would be happy to provide a working draft of any manuscript currently in preparation.